

Bibliotek for å lese og arbeide med OSM-data

Dokumentasjon

Gruppe 8

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Dato: 27. mai 2025

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1 Introduksjon

OsmToolkit er et .NET-bibliotek for lesing og skriving, men også søking, analysering og filtrering av geografiske data fra *OpenStreetMap* (*OSM*). Biblioteket fokuserer på støtte for Json og XML-baserte .osm-filer og tilbyr et brukervennlig API for arbeid med noder, veier og relasjoner representert gjennom den datamodellen *OsmData*.

1.1 Hva gjør biblioteket og hvem er det for?

1.1.1 Med OsmToolkit kan du:

- Lese og tolke OSM-data direkte fra XML- eller Json-filer, tekststrenger eller streams, og enkelt gjøre dem tilgjengelig som strukturert C# data gjennom vår interne modell.
- Skrive og eksportere OSM-data tilbake til XML eller Json, for eksempel etter redigering, filtrering eller analyse.
- Utføre avanserte søk og analyser. Blant annet filtrere OSM-objekter basert på tags, vei og distanse. Regne ut korteste strekning mellom to punkter, eller søke etter OSM-objekter innenfor en gitt radius målt i luftlinje fra koordinater. Veisøk med fleksible søke- og filteringsfunksjoner.
- Bygge ut og tilpasse etter egne behov. Man har gode muligheter for å utvide funksjonalitet med egne søk, nye filformater eller spesialtilpassede analyser.
- Jobbe effektivt med asynkrone operasjoner. Alle våres lese og skriveoperasjoner er asynkrone for å sikre høy ytelse og god brukeropplevelse i moderne apper.
- Biblioteket har innebygd støtte for logging via *ILogger<T>*, og er utformet for enkel integrasjon med Dependency Injection. Dersom ingen logger injiseres, benyttes automatisk en *NullLogger* for å sikre stabil oppførsel.

1.1.2 Hvem er biblioteket for?

OsmToolkit er laget for de som ønsker å jobbe profesjonelt og effektivt med OSM-data i .NET. Typiske brukere for dette biblioteket er:

- Utviklere som lager kartapplikasjoner, geografisk visualisering, logistikk eller analyseverktøy der OSM-data spiller en nøkkelrolle.
- Bedrifter og studenter som trenger en pålitelig og utvidbart rammeverk for å integrere kartdata samt geografisk analysere egne løsninger.
- Forskere og dataanalytikere som ønsker å kunne behandle og analysere større mengder med OSM-data i et .NET-miljø.

2 Installasjon og oppsett

2.1 Steg-for-steg: Installere og bruke biblioteket

OsmToolkit distribueres som en NuGet-pakke og kan enkelt legges til i ethvert .NET-prosjekt.

1. Opprett et nytt .NET-prosjekt (for eksempel en konsollapplikasjon eller et API-prosjekt).
2. Legg til nuget pakken OsmToolkit.nupkg via nuget manager i Visual Studio
3. Bekreft at prosjektet er konfigurert med .NET 8 SDK.
4. Registrer tjenestene i Dependency Injection-containeren:

```
using OsmToolkit;  
using Microsoft.Extensions.DependencyInjection;  
  
var services = new ServiceCollection();  
services.AddOsmToolkit();
```

5. Du kan nå bruke bibliotekets funksjonalitet, for eksempel:

```
var deserializer = services.BuildServiceProvider()  
    .GetRequiredService<IOsmDeserializer>();  
  
var data = await deserializer.DeserializeFromFileAsync("map.osm");
```

2.2 Avhengigheter

- **Krav:**
 - .NET 8 SDK
- **Automatisk avhengighet:** Microsoft.Extensions.Logging.Abstractions brukes for intern logging og installeres automatisk som del av NuGet-pakken.
- **Dependency Injection:** Biblioteket er laget med støtte for DI via IServiceCollection. Alle nødvendige tjenester registreres med services.AddOsmToolkit();.

3 Overordnet struktur

3.1 Modulær oppbygging

Biblioteket er delt inn i flere hovedmoduler:

- **Modeller:** Inneholder klasser som definerer OSM-data. Eksempler er `OsmEntity`, `Node`, `Way`, `Relation`, samt tilhørende klasser som `User` og enums for `ReferenceType`.
- **Finder:** Inneholder `IOsmFinderV2` for alle Finder metoder, men inneholder også spesifikke interfaces for enkelte metodegrupper med implementasjon for standard søkefunksjon til å finne `OsmEntity` instanser med spesifikke tags eller innenfor visse rekkevidder.
- **Serializer:** Inneholder grensesnittene `IOsmSerializer` og `IOsmDeserializer` med implementasjoner for XML og Json basert serialisering og deserialisering.

3.2 Sammenheng mellom modulene

De ulike modulene i `OsmToolkit` er designet for å være selvstendige, men samarbeider tett gjennom veldefinerte grensesnitt og avhengigheter. Dette gir et fleksibelt og utvidbart system, hvor hver modul har sitt tydelige ansvar. Slik henger modulene sammen:

- **Models** utgjør grunnmuren og definerer alle sentrale datastrukturer (for eksempel `Node`, `Way`, `Relation`). Alle andre moduler bruker disse modellene for å lese, skrive og manipulere OSM-data.
- **Serialization**-modulen sørger for at data fra **Models** kan lagres til og hentes fra eksterne kilder, som XML eller JSON filer. Når man deserialiserer en fil, opprettes objekter i **Models**-modulen. Serialisering og deserialisering utføres gjennom grensesnitt som `IOsmSerializer` og `IOsmDeserializer`.
- **Finders** bruker modellene fra **Models** til å utføre søk og filtrering. De er koblet til datamodellene via flere grensesnitt som `IOsmFinder` og `IOsmFinderV2` (eller spesifikke Finder-interfaces for enkelte metodegrupper), og returnerer alltid objekter av typer som er definert i **Models**. Finders modulen kan også bruke alternativer fra **Options** for å tilpasse søk.
- **ServiceCollectionExtensions** gjør det enkelt å koble sammen alle de andre modulene ved å registrere nødvendige tjenester i Dependency Injection-containeren. Dette sikrer at komponentene blir riktig instansiert og at avhengigheter mellom dem håndteres automatisk.

Kort forklart: Modellen vår gir tydelig ansvarsdeling: Data defineres i **Models**, håndteres og lagres via **Serialization**, søkes gjennom **Finders**, og hele biblioteket kobles sammen med **ServiceCollectionExtensions**. Dette gjør at hver modul kan videreutvikles eller byttes ut uavhengig av resten, så lenge de overholder felles grensesnitt.

3.3 Feilhåndtering

Robust og forutsigbar feilhåndtering var viktig for oss i utviklingen av biblioteket. Vi følger et prinsipp som kalles 'fail fast', der metoder validerer input tidlig og kaster tydelige exceptions ved feil eller ugyldige data. Dette gir forutsigbar oppførsel og gjør det enkelt å fange opp og håndtere feil.

Validering av input:

- Ved deserialisering fra fil, streng eller stream, kontrolleres alltid filendelsen. Kun `.osm` og `.xml` godtas for XML-deserialisering, og `.json` for JSON-deserialisering.
- Feil filtype gir en `ArgumentException` med forklarende melding.
- Manglende eller feil formaterte attributter kaster en `InvalidDataException`.
- Generelle parsing-feil håndteres med `FormatException`.

Feilhåndtering under IO-operasjoner:

- Alle metoder som leser fra eller skriver til filer og streams håndterer vanlige IO-feil.
- Vi benytter trygge `FileStream`- og `StreamReader`-konstruksjoner slik at systemfeil som `FileNotFoundException`, `IOException`, `UnauthorizedAccessException` og `PathTooLongException` kastes videre.
- Asynkrone metoder for inn og utlasting av data støtter `CancellationToken` for trygg avbruddshåndtering.
- Feil og unntak som oppstår i lavere lag (f.eks. filtilgang eller ugyldig stream) bobler opp, slik at man kan fange dem i `try/catch`-blokker.

3.4 Logging

Formål:

Logging er sentralt i biblioteket for feilsøking, ytelsesmåling og sporing av hva som skjer under serialisering og deserialisering. Vi benytter Microsoft.Extensions.Logging sitt `ILogger<T>`-interface gjennomgående. Dette gir muligheten til å velge logger (for eksempel `ConsoleLogger`, fil-logger eller eksterne tjenester), uten å gjøre endringer i biblioteket.

Konfigurasjon:

Loggeren settes opp via dependency injection. Dersom man ikke eksplisitt konfigurerer logger, benytter biblioteket `NullLogger`, slik at logging aldri fører til unødvendige feil.

Eksempel på oppsett av `ConsoleLogger`:

```
var services = new ServiceCollection();
services.AddLogging(builder => builder.AddConsole());
services.AddOsmToolkit();
```

Med dette oppsettet vil alle loggmeldinger fra biblioteket vises i konsollen. Andre loggtyper (fil, database, ekstern tjeneste) kan benyttes ved å endre loggerkonfigurasjonen uten behov for endringer i selve biblioteket.

4 Grunnleggende bruk

Grunnleggende bruk av biblioteket med noen få eksempler på bruksområder.

4.1 Eksempel på bruk av biblioteket

Under ligger noen eksempler til hvordan man kan bruke de ulike delene av biblioteket, full liste over metoder ligger lenger ned i dokumentet.

Listing 1: Eksempel på bruk av bare deserializer på .osm fil.

```
var services = new ServiceCollection();
services.AddOsmToolkit();

var deserializer = services.BuildServiceProvider()
    .GetRequiredService<IOsmXmlDeserializer>();

var data = await deserializer.DeserializeFromFileAsync("map.osm");

foreach (var item in data.Nodes)
{
    Console.WriteLine($"node id: '{item.Id.ToString()}'");
    foreach (var tag in item.Tags)
    {
        Console.WriteLine($"Tag: {tag}");
    }
}
```

Listing 2: Eksempel på bruk finder.

```
var finder = services.BuildServiceProvider()
    .GetRequiredService<IOsmFinder<OsmEntity>>();

var result = finder.FindByTag(data, "amenity");

foreach (var node in result.Nodes)
{
    Console.WriteLine($"Resultat: {node.Id}, [{node.Latitude},
        {node.Longitude}]");
}
```

Listing 3: Eksempel på bruk av serializer

```
var serializer = services.BuildServiceProvider()
    .GetRequiredService<IOsmXmlSerializer>();
await serializer.SerializeToFileAsync(data, "map.osm");
```

Listing 4: Eksempel på å finne alle kafeer ved hjelp av tags

```
var services = new ServiceCollection();
services.AddOsmToolkit();

var deserializer = services.BuildServiceProvider()
    .GetRequiredService<IOsmXmlDeserializer>();

var data = await deserializer.DeserializeFromFileAsync("map.osm");

// Finn alle noder med taggen amenity=cafe
var cafes = data.Nodes.Where(HasAnyTag([("amenity", "cafe")]));

foreach (var cafe in cafes)
{
    if (cafe.HasAnyTagKey("name"))
    {
        var name = cafe.Tags.GetValueOrDefault("name");
        Console.WriteLine($"Kafe: {name} (ID: {cafe.Id})");
    }
    else
        Console.WriteLine($"Kafe uten navn (ID: {cafe.Id})");
}
```

Kommentar: Her settes biblioteket opp med Dependency Injection, data deserialiseres fra fil, og kafeer hentes ut ved hjelp av filtrering på tags.

Listing 5: Eksempel på logging av tomme relasjoner

```
var services = new ServiceCollection();
services.AddOsmToolkit();

var serviceProvider = services.BuildServiceProvider();
var deserializer =
    serviceProvider.GetRequiredService<IOsmXmlDeserializer>();
var logger = serviceProvider.GetRequiredService<ILogger<OsmData>>();

var data = await deserializer.DeserializeFromFileAsync("map.osm");

foreach (var relation in data.Relations)
{
    if (relation.Members.Count == 0)
        logger.LogWarning($"Relation {relation.Id} is empty.");
}
```

Kommentar: Viser hvordan man logger alle relasjoner uten medlemmer. Loggeren konfigureres automatisk, så koden virker både med og uten eksplisitt logging i prosjektet.

Listing 6: Eksempel på asynkron lesing av flere filer samtidig

```
var services = new ServiceCollection();
services.AddOsmToolkit();

var deserializer = services.BuildServiceProvider()
    .GetRequiredService<IOsmXmlDeserializer>();

var files = Directory.GetFiles("maps", "*.osm");

var tasks = files.Select(async file =>
{
    var localData = await deserializer.DeserializeFromFileAsync(file);
    Console.WriteLine($"{file}: {localData.Nodes.Count} nodes loaded.");
});
await Task.WhenAll(tasks);
```

Kommentar: Her ser du hvordan biblioteket enkelt kan brukes til å lese mange filer parallelt med asynkronne operasjoner, for eksempel ved batchprosessering.

Listing 7: Eksempel på å finne korteste vei mellom to noder

```
var services = new ServiceCollection();
services.AddOsmToolkit();

var deserializer = services.BuildServiceProvider()
    .GetRequiredService<IOsmXmlDeserializer>();

var data = await deserializer.DeserializeFromFileAsync("map.osm");

var finder = services.BuildServiceProvider()
    .GetRequiredService<IOsmFinderV2>();

// Finn korteste vei
var pathOptions = new PathOptions(TravelMode.MotorCar, false)
var path = finder.FindShortestPath(data, data.nodes[10], data.nodes[58],
    pathOptions);

foreach (var node in path.Data.Nodes)
{
    Console.WriteLine($"{node.Latitude.ToString(CultureInfo.InvariantCulture)},
        {node.Longitude.ToString(CultureInfo.InvariantCulture)}");
}
```

Kommentar: Her prøver den å finne den korteste veien mellom to tilfeldige noder i datasettet, med bil som reisemetode og motorveier tillatt. Nodenes koordinater blir printet ut slik at du kan visuelt legge punktene inn i <https://www.mapcustomizer.com/#>.

A API-referanse

For klikkbare linker, se egen inkludert PDF fil.

Namespace OsmToolkit

Classes

[Member](#)

A [Member](#) within a [Relation](#) instance.

[Node](#)

Represents a point of interest on earth.

[OsmCoordinateBounds](#)

Represents the latitude and longitude bounds for OSM-data.

[OsmData](#)

Contains lists including all [Node](#), [Way](#) and [Relation](#) instances registered from OSM-sources.

[OsmEntity](#)

Represents a map element.

[OsmHeader](#)

Represents the header of the OSM-data, describing where it was generated and information about copyright.

[OsmPath](#)

Represents a path of multiple [OsmEntity](#) instances between two [Node](#) instances.

[Relation](#)

Representation of a geographic feature.

[ServiceCollectionExtensions](#)

[User](#)

Contains user data about OSM-creators.

[Way](#)

Connection between two or more nodes.

Enums

[ReferenceType](#)

Contains [OsmEntity](#) reference types.

Class Member

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

A [Member](#) within a [Relation](#) instance.

```
public class Member
```

Inheritance

[object](#)  ← Member

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Member(ReferenceType, long)

Initializes a new [Member](#) object without a specified role.

```
public Member(ReferenceType type, long referenceId)
```

Parameters

type [ReferenceType](#)

The specified [OsmEntity](#) type.

referenceId [long](#) 

The reference to the id of [OsmEntity](#), must be of values above 0.

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if `referenceId` is less than or equal to 0.

Member(ReferenceType, long, string?)

Initializes a new [Member](#) object with a specified role.

```
public Member(ReferenceType type, long referenceId, string? role)
```

Parameters

type [ReferenceType](#)

The specified [OsmEntity](#) type.

referenceId [long](#)[↗]

The reference to the id of [OsmEntity](#), must be of values above 0.

role [string](#)[↗]

The specified [Member](#) role, can be null.

Exceptions

[ArgumentOutOfRangeException](#)[↗]

Throws if `referenceId` is less than or equal to 0.

Properties

ReferenceId

A reference to the id of the [OsmEntity](#) instance.

```
[JsonPropertyName("ref")]  
[JsonPropertyOrder(1)]  
public long ReferenceId { get; }
```

Property Value

[long](#)[↗]

Role

The specified [Member](#) role for a [Relation](#).

```
[JsonPropertyName("role")]
[JsonPropertyOrder(2)]
public string Role { get; set; }
```

Property Value

[string](#)

Type

The specified [OsmEntity](#) type.

```
[JsonPropertyName("type")]
[JsonPropertyOrder(0)]
public ReferenceType Type { get; set; }
```

Property Value

[ReferenceType](#)

Class Node

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents a point of interest on earth.

```
public class Node : OsmEntity
```

Inheritance

[object](#)  ← [OsmEntity](#) ← Node

Inherited Members

[OsmEntity.Tags](#) , [OsmEntity.Id](#) , [OsmEntity.Visible](#) , [OsmEntity.Version](#) , [OsmEntity.ChangeSet](#) ,
[OsmEntity.Timestamp](#) , [OsmEntity.User](#) , [OsmEntity.AddTag\(string, string\)](#) , [OsmEntity.RemoveTag\(string\)](#) ,
[OsmEntity.HasTagKey\(string\)](#) , [OsmEntity.HasAnyTagKey\(IEnumerable<string>\)](#) ,
[OsmEntity.HasAllTagKeys\(IEnumerable<string>\)](#) , [OsmEntity.HasTagValue\(string\)](#) ,
[OsmEntity.HasAnyTagValue\(IEnumerable<string>\)](#) , [OsmEntity.HasAllTagValues\(IEnumerable<string>\)](#) ,
[OsmEntity.HasTag\(string, string\)](#) , [OsmEntity.HasAnyTag\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAnyTag\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAnyTag\(IEnumerable<KeyValuePair<string, string>>\)](#) ,
[OsmEntity.HasAllTags\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAllTags\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAllTags\(IEnumerable<KeyValuePair<string, string>>\)](#) , [object.Equals\(object\)](#)  ,
[object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Node(long, bool, int, long, DateTime, User, double, double)

Initializes a new [Node](#) object without specified tags.

```
public Node(long id, bool visible, int version, long changeSet, DateTime timestamp, User  
user, double latitude, double longitude)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **latitude** is out of the range -90 to 90 or **longitude** is out of the range -180 to 180.

Node(long, bool, int, long, DateTime, User, double, double, Dictionary<string, string>?)

Initializes a new [Node](#) object with specified tags.

[JsonConstructor]

```
public Node(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, double latitude, double longitude, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id`, `version` or `changeSet` is less than or equal to 0, or if `latitude` is out of the range -90 to 90 or `longitude` is out of the range -180 to 180.

[ArgumentNullException](#)

Throws if `user` is `null`.

Node(long, double, double)

Initializes a new base [Node](#) without specified tags.

```
public Node(long id, double latitude, double longitude)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`latitude` [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

`longitude` [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `latitude` is out of the range -90 to 90 or `longitude` is out of the range -180 to 180.

Node(long, double, double, Dictionary<string, string>?)

Initializes a new base [Node](#) with specified tags.

```
public Node(long id, double latitude, double longitude, Dictionary<string, string>? tags
= null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **latitude** is out of the range -90 to 90 or **longitude** is out of the range -180 to 180.

Properties

Latitude

Distance north or south of equator, within the range of -90 to 90.

```
[JsonPropertyName("lat")]
[JsonPropertyOrder(6)]
public double Latitude { get; }
```

Property Value

[double](#)

Longitude

Distance east or west of prime meridian, within the range of -180 to 180.

```
[JsonPropertyName("lon")]
[JsonPropertyOrder(7)]
public double Longitude { get; }
```

Property Value

[double](#)

Methods

IsPartOf(Relation)

Checks if [Node](#) is part of a [Relation](#) instance.

```
public bool IsPartOf(Relation relation)
```

Parameters

relation [Relation](#)

[Relation](#) to check if [Node](#) is part of.

Returns

[bool](#)

true if [Node](#) is part of **relation**. Otherwise, **false**.

Exceptions

[ArgumentNullException](#)

Throws if **relation** is **null**.

IsPartOf(Way)

Checks if [Node](#) is part of a [Way](#) instance.

```
public bool IsPartOf(Way way)
```

Parameters

way [Way](#)

[Way](#) instance to check if [Node](#) is part of.

Returns

[bool](#)[↗]

`true` if [Node](#) is part of `way`. Otherwise, `false`.

Exceptions

[ArgumentNullException](#)[↗]

Throws if `way` is `null`.

IsPartOf(Way, Relation)

Checks if [Node](#) is part of both a [Way](#) instance and a [Relation](#) instance.

```
public bool IsPartOf(Way way, Relation relation)
```

Parameters

way [Way](#)

[Way](#) to check if [Node](#) is part of.

relation [Relation](#)

[Relation](#) to check if [Node](#) is part of.

Returns

[bool](#)↗

true if [Node](#) is part of way and relation. Otherwise, false.

Exceptions

[ArgumentNullException](#)↗

Throws if way or relation is null.

Class OsmCoordinateBounds

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents the latitude and longitude bounds for OSM-data.

```
public class OsmCoordinateBounds
```

Inheritance

[object](#) ← OsmCoordinateBounds

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

OsmCoordinateBounds(double, double, double, double)

Initializes a new instance of the [OsmCoordinateBounds](#) class.

```
public OsmCoordinateBounds(double minimumLatitude, double minimumLongitude, double maximumLatitude, double maximumLongitude)
```

Parameters

minimumLatitude [double](#)

Minimum distance north or south of equator, within the range of -90 to 90.

minimumLongitude [double](#)

Minimum distance east or west of prime meridian, within the range of -180 to 180.

maximumLatitude [double](#)

Maximum distance north or south of equator, within the range of -90 to 90.

maximumLongitude [double](#)

Maximum distance east or west of prime meridian, within the range of -180 to 180.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `minimumLatitude` or `minimumLatitude` is out of the range -90 to 90, `minimumLongitude` or `maximumLongitude` is out of the range -180 to 180, `minimumLatitude` is equal to or greater than `maximumLatitude` or `minimumLongitude` is equal to or greater than `maximumLongitude`.

Properties

MaximumLatitude

Maximum distance north or south of equator, within the range of -90 to 90.

```
[JsonPropertyName("maxlat")]
[JsonPropertyOrder(2)]
public double MaximumLatitude { get; }
```

Property Value

[double](#)

MaximumLongitude

Maximum distance east or west of prime meridian, within the range of -180 to 180.

```
[JsonPropertyName("maxlon")]
[JsonPropertyOrder(3)]
public double MaximumLongitude { get; }
```

Property Value

[double](#)

MinimumLatitude

Minimum distance north or south of equator, within the range of -90 to 90.

```
[JsonPropertyName("minlat")]  
[JsonPropertyOrder(0)]  
public double MinimumLatitude { get; }
```

Property Value

[double](#)

MinimumLongitude

Minimum distance east or west of prime meridian, within the range of -180 to 180.

```
[JsonPropertyName("minlon")]  
[JsonPropertyOrder(1)]  
public double MinimumLongitude { get; }
```

Property Value

[double](#)

Class OsmData

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Contains lists including all [Node](#), [Way](#) and [Relation](#) instances registered from OSM-sources.

```
public class OsmData
```

Inheritance

[object](#)  ← OsmData

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

OsmData(OsmHeader)

Initializes a new [OsmData](#) object without any registered instances of [OsmEntity](#).

```
public OsmData(OsmHeader header)
```

Parameters

header [OsmHeader](#)

The [OsmHeader](#) describing the header details for the OSM-data.

Exceptions

[ArgumentNullException](#) 

Throws if **header** is `null`.

OsmData(OsmHeader, OsmCoordinateBounds?, IEnumerable<Node>?, IEnumerable<Way>?, IEnumerable<Relation>?)

Initializes a new [OsmData](#) object with registered instances of class types [Node](#), [Way](#) and [Relation](#).

```
public OsmData(OsmHeader header, OsmCoordinateBounds? bounds = null, IEnumerable<Node>? nodes = null, IEnumerable<Way>? ways = null, IEnumerable<Relation>? relations = null)
```

Parameters

header [OsmHeader](#)

The [OsmHeader](#) describing the header details for the OSM-data.

bounds [OsmCoordinateBounds](#)

The [OsmCoordinateBounds](#) set for the OSM-data, can be `null`.

nodes [IEnumerable](#) [Node](#)

List of [Node](#) instances, can be `null`.

ways [IEnumerable](#) [Way](#)

List of [Way](#) instances, can be `null`.

relations [IEnumerable](#) [Relation](#)

List of [Relation](#) instances, can be `null`.

Exceptions

[ArgumentNullException](#)

Throws if **header** is `null`.

Properties

Bounds

[OsmCoordinateBounds](#) set for the OSM-data.

```
[JsonPropertyName("bounds")]
[JsonPropertyOrder(1)]
public OsmCoordinateBounds? Bounds { get; }
```

Property Value

[OsmCoordinateBounds](#)

Header

[OsmHeader](#) describing header details for the OSM-data.

```
[JsonPropertyName("header")]
[JsonPropertyOrder(0)]
public OsmHeader Header { get; set; }
```

Property Value

[OsmHeader](#)

Nodes

List of registered [Node](#) instances.

```
[JsonPropertyName("nodes")]
[JsonPropertyOrder(2)]
public IList<Node> Nodes { get; }
```

Property Value

[IList](#) [<Node>](#)

Relations

List of registered [Relation](#) instances.

```
[JsonPropertyName("relations")]
[JsonPropertyOrder(4)]
public IList<Relation> Relations { get; }
```

Property Value

[IList](#) [<Relation>](#)

Ways

List of registered [Way](#) instances.

```
[JsonPropertyName("ways")]
[JsonPropertyOrder(3)]
public IList<Way> Ways { get; }
```

Property Value

[IList](#) [<Way>](#)

Methods

SortOsmData()

Sorts all OSM entities by their unique identifiers (Id) to ensure consistent serialization order.

```
public void SortOsmData()
```

Class OsmEntity

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents a map element.

```
public abstract class OsmEntity
```

Inheritance

[object](#)  ← OsmEntity

Derived

[Node](#), [Relation](#), [Way](#)

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

OsmEntity(long)

Initializes a new default [OsmEntity](#) object with just a specified id.

```
protected OsmEntity(long id)
```

Parameters

id [long](#) 

Unique identifier for [OsmEntity](#), must be of values above 0.

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if **id** is less than or equal to 0.

OsmEntity(long, bool, int, long, DateTime)

Initializes a new [OsmEntity](#) object with no tags.

```
protected OsmEntity(long id, bool visible, int version, long changeSet, DateTime timestamp)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#), must be of values above 0.

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active, **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0.

OsmEntity(long, bool, int, long, DateTime, User?, Dictionary<string, string>?)

Initializes a new [OsmEntity](#) object with tags.

```
protected OsmEntity(long id, bool visible, int version, long changeSet, DateTime timestamp, User? user = null, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#), must be of values above 0.

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0.

OsmEntity(long, Dictionary<string, string>?)

Intializes a new default [OsmEntity](#) object with just a specified id and tags.

```
protected OsmEntity(long id, Dictionary<string, string>? tags = null)
```


Parameters

id [long](#)

Unique identifier for [OsmEntity](#), must be of values above 0.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0.

Properties

ChangeSet

Id of the set of changes the [OsmEntity](#) was last modified in.

```
[JsonPropertyName("changeset")]
[JsonPropertyOrder(3)]
public long ChangeSet { get; set; }
```

Property Value

[long](#)

Id

Unique identifier for [OsmEntity](#).

```
[JsonPropertyName("id")]
[JsonPropertyOrder(0)]
public long Id { get; }
```

Property Value

[long](#)

Tags

A read-only dictionary of key-value string pairs representing metadata for [OsmEntity](#).

```
[JsonPropertyName("tags")]
[JsonPropertyOrder(8)]
public IReadOnlyDictionary<string, string> Tags { get; }
```

Property Value

[IReadOnlyDictionary](#) <[string](#), [string](#)>

Timestamp

Last time [OsmEntity](#) was changed.

```
[JsonPropertyName("timestamp")]
[JsonPropertyOrder(4)]
public DateTime Timestamp { get; set; }
```

Property Value

[DateTime](#)

User

Creator of [OsmEntity](#).

```
[JsonPropertyName("user")]
[JsonPropertyOrder(5)]
public User? User { get; set; }
```

Property Value

[User](#)

Version

Current version number of [OsmEntity](#).

```
[JsonPropertyName("version")]
[JsonPropertyOrder(2)]
public int Version { get; set; }
```

Property Value

[int](#)

Visible

Indicates whether the [OsmEntity](#) is currently visible or active, **true** if it is still part of the active map data, or **false** if otherwise.

```
[JsonPropertyName("visible")]
[JsonPropertyOrder(1)]
public bool Visible { get; set; }
```

Property Value

[bool](#)

Methods

AddTag(string, string)

Adds or updates a dictionary containing metadata for [OsmEntity](#).

```
public void AddTag(string key, string value)
```

Parameters

key [string](#)

The key value to access the corresponding metadata, cannot be `null` or an empty string.

value [string](#)

The value describing the corresponding metadata, if `null` it will be set as an empty string.

Exceptions

[ArgumentException](#)

Throws if `key` is `null` or an empty string.

HasAllTagKeys(IEnumerable<string>)

Checks if the [OsmEntity](#) has tags with all of the specified `keys`.

```
public bool HasAllTagKeys(IEnumerable<string> keys)
```

Parameters

`keys` [IEnumerable](#) <[string](#)>

Keys to look for.

Returns

[bool](#)

`true`, if tags with all of the `keys` were found. Otherwise; `false`.

HasAllTagValues(IEnumerable<string>)

Checks if the [OsmEntity](#) has tags with all of the specified `values`.

```
public bool HasAllTagValues(IEnumerable<string> values)
```

Parameters

values [IEnumerable](#) <[string](#)>

Values to look for.

Returns

[bool](#)

true, if tags with all of the **values** were found. Otherwise; **false**.

HasAllTags(Dictionary<string, string>)

Checks if the [OsmEntity](#) has tags matching all of the specified **tags**.

```
public bool HasAllTags(Dictionary<string, string> tags)
```

Parameters

tags [Dictionary](#) <[string](#), [string](#)>

Tags to look for.

Returns

[bool](#)

true, if tags matching all of the **tags** were found. Otherwise; **false**.

HasAllTags(IEnumerable<KeyValuePair<string, string>>)

Checks if the [OsmEntity](#) has tags matching all of the specified **tags**.

```
public bool HasAllTags(IEnumerable<KeyValuePair<string, string>> tags)
```

Parameters

tags [IEnumerable](#) <[KeyValuePair](#) <[string](#), [string](#)>>

Tags to look for.

Returns

[bool](#)

`true`, if tags matching all of the `tags` were found. Otherwise; `false`.

HasAllTags(params (string key, string value)[])

Checks if the [OsmEntity](#) has tags matching all of the specified `tags`.

```
public bool HasAllTags(params (string key, string value)[] tags)
```

Parameters

`tags` ([string](#) `key`, [string](#) `value`)[]

Tags to look for.

Returns

[bool](#)

`true`, if tags matching all of the `tags` were found. Otherwise; `false`.

HasAnyTag(Dictionary<string, string>)

Checks if the [OsmEntity](#) has any tag matching one the specified `tags`.

```
public bool HasAnyTag(Dictionary<string, string> tags)
```

Parameters

`tags` [Dictionary](#) `<string, string>`

Tags to look for.

Returns

[bool](#)

`true`, if a tag with matches one of the `tags` was found. Otherwise; `false`.

HasAnyTag(IEnumerable<KeyValuePair<string, string>>)

Checks if the [OsmEntity](#) has any tag matching one the specified `tags`.

```
public bool HasAnyTag(IEnumerable<KeyValuePair<string, string>> tags)
```

Parameters

`tags` [IEnumerable](#) <[KeyValuePair](#) <[string](#), [string](#)> >

Tags to look for.

Returns

[bool](#)

`true`, if a tag with matches one of the `tags` was found. Otherwise; `false`.

HasAnyTag(params (string key, string value)[])

Checks if the [OsmEntity](#) has any tag matching one the specified `tags`.

```
public bool HasAnyTag(params (string key, string value)[] tags)
```

Parameters

`tags` ([string](#) `key`, [string](#) `value`)[]

Tags to look for.

Returns

[bool](#)

`true`, if a tag with matches one of the `tags` was found. Otherwise; `false`.

HasAnyTagKey(IEnumerable<string>)

Checks if the [OsmEntity](#) has a tag with any key of the specified **keys**.

```
public bool HasAnyTagKey(IEnumerable<string> keys)
```

Parameters

keys [IEnumerable](#) <[string](#)>

Keys to look for.

Returns

[bool](#)

true, if a tag with any of the **keys** was found. Otherwise; **false**.

HasAnyTagValue(IEnumerable<string>)

Checks if the [OsmEntity](#) has a tag with any value of the specified **values**.

```
public bool HasAnyTagValue(IEnumerable<string> values)
```

Parameters

values [IEnumerable](#) <[string](#)>

Values to look for.

Returns

[bool](#)

true, if a tag with any of the **values** was found. Otherwise; **false**.

HasTag(string, string)

Checks if the [OsmEntity](#) has a tag matching the specified key and value.


```
public bool HasTag(string key, string value)
```

Parameters

key [string](#)

Key to look for.

value [string](#)

Value to look for.

Returns

[bool](#)

true, if a tag with matches with **key** and **value** was found. Otherwise; **false**.

HasTagKey(string)

Checks if the [OsmEntity](#) has a tag with the specified **key**.

```
public bool HasTagKey(string key)
```

Parameters

key [string](#)

Key to look for.

Returns

[bool](#)

true, if a tag with **key** was found. Otherwise; **false**.

HasTagValue(string)

Checks if the [OsmEntity](#) has a tag with the specified **value**.

```
public bool HasTagValue(string value)
```

Parameters

value [string](#)

Value to look for.

Returns

[bool](#)

`true`, if a tag with `value` was found. Otherwise; `false`.

RemoveTag(string)

Removes a single dictionary containing metadata for [OsmEntity](#).

```
public bool RemoveTag(string key)
```

Parameters

key [string](#)

The key value to access the corresponding metadata, cannot be `null` or an empty string.

Returns

[bool](#)

`true` if removal is successful; otherwise, `false`.

Class OsmHeader

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents the header of the OSM-data, describing where it was generated and information about copyright.

```
public class OsmHeader
```

Inheritance

[object](#)  ← OsmHeader

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

OsmHeader(double)

Initiliazes a new default [OsmHeader](#) with version.

```
public OsmHeader(double version)
```

Parameters

version [double](#) 

Current version number of the OSM-data. Cannot be less than or equal to 0

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if **version** is less than or equal to 0.

OsmHeader(double, string?, string?, string?, string?)

Initiliazes a new [OsmHeader](#) object.

```
public OsmHeader(double version, string? generator, string? copyright, string? attributionUrl, string? licenseUrl)
```

Parameters

version [double](#) 

Current version number of the OSM-data. Cannot be less than or equal to 0

generator [string](#) 

Generator used to create the OSM-data. If **null**, it will be set to "".

copyright [string](#) 

Copyright owners of the OSM-data. If **null**, it will be set to "".

attributionUrl [string](#) 

URL to the copyright owners' copyright page. If **null**, it will be set to "".

licenseUrl [string](#) 

URL to the copyright owners' license of use. If **null**, it will be set to "".

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if **version** is less than or equal to 0.

Properties

AttributionUrl

Contains an URL to the copyright owners' copyright page.

```
[JsonPropertyName("attribution")]
[JsonPropertyOrder(3)]
public string AttributionUrl { get; }
```

Property Value

[string](#) 

Copyright

Describes the copyright owners of the OSM-data.

```
[JsonPropertyName("copyright")]
[JsonPropertyOrder(2)]
public string Copyright { get; }
```

Property Value

[string](#) 

Generator

Describes the generator used to create the OSM-data.

```
[JsonPropertyName("generator")]
[JsonPropertyOrder(1)]
public string Generator { get; }
```

Property Value

[string](#) 

LicenseUrl

Contains an URL to the copyright owners' license of use.

```
[JsonPropertyName("license")]  
[JsonPropertyOrder(4)]  
public string LicenseUrl { get; }
```

Property Value

[string](#)

Version

Current version number of the OSM schema.

```
[JsonPropertyName("version")]  
[JsonPropertyOrder(0)]  
public double Version { get; }
```

Property Value

[double](#)

Class OsmPath

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents a path of multiple [OsmEntity](#) instances between two [Node](#) instances.

```
public class OsmPath
```

Inheritance

[object](#)  ← OsmPath

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

OsmPath(OsmData, double)

Initializes a new and empty [OsmPath](#) object with no description.

```
public OsmPath(OsmData data, double totalDistance)
```

Parameters

data [OsmData](#)

Data containing [OsmEntity](#) instances making up the path.

totalDistance [double](#) 

Total distance of the path in meters.

OsmPath(OsmData, double, Node?, Node?)

Initializes a new [OsmPath](#) object with a startnode and endnode, but no description.

```
public OsmPath(OsmData data, double totalDistance, Node? startNode = null, Node? endNode = null)
```

Parameters

data [OsmData](#)

Data containing [OsmEntity](#) instances making up [OsmPath](#).

totalDistance [double](#)[↗]

Total distance of [OsmPath](#) in meters.

startNode [Node](#)

Startpoint of [OsmPath](#), must already be in **data** and a [Way](#), otherwise it is set to **null**.

endNode [Node](#)

Endpoint of [OsmPath](#), must already be in **data** and a [Way](#), otherwise it is set to **null**

OsmPath(OsmData, double, Node?, Node?, string?)

Initializes a new [OsmPath](#) object with a description, startnode and endnode.

```
public OsmPath(OsmData data, double totalDistance, Node? startNode = null, Node? endNode = null, string? description = null)
```

Parameters

data [OsmData](#)

Data containing [OsmEntity](#) instances making up [OsmPath](#).

totalDistance [double](#)[↗]

Total distance of [OsmPath](#) in meters.

startNode [Node](#)

Startpoint of [OsmPath](#), must already be in **data** and a [Way](#), otherwise it is set to **null**.

endNode [Node](#)

Endpoint of [OsmPath](#), must already be in **data** and a [Way](#), otherwise it is set to **null**.

description [string](#)↗

Optional description for additional information or messages.

OsmPath(OsmData, double, string?)

Initializes a new and empty [OsmPath](#) object with a description.

```
public OsmPath(OsmData data, double totalDistance, string? description = null)
```

Parameters

data [OsmData](#)

Data containing [OsmEntity](#) instances making up [OsmPath](#).

totalDistance [double](#)↗

Total distance of [OsmPath](#) in meters.

description [string](#)↗

Optional description for additional information or messages.

Properties

Data

Data containing all of the [OsmEntity](#) instances making up [OsmPath](#).

```
public OsmData Data { get; }
```

Property Value

[OsmData](#)

Description

Optional description containing additional information or messages when necessary.

```
public String? Description { get; }
```

Property Value

[string](#)[↗]

EndNode

[Node](#) instance representing the **endpoint** of [OsmPath](#), is **null** if path is empty.

```
public Node? EndNode { get; }
```

Property Value

[Node](#)

StartNode

[Node](#) instance representing the **startpoint** of [OsmPath](#), is **null** if path is empty.

```
public Node? StartNode { get; }
```

Property Value

[Node](#)

TotalDistance

Total distance of the path in meters, is 0 if path is empty.

```
public double TotalDistance { get; }
```

Property Value

[double](#) 

Enum ReferenceType

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Contains [OsmEntity](#) reference types.

```
public enum ReferenceType
```

Fields

```
node = 0
```

Represents members of class type [Node](#).

```
relation = 2
```

Represents members of class type [Relation](#).

```
way = 1
```

Represents members of class type [Way](#).

Class Relation

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Representation of a geographic feature.

```
public class Relation : OsmEntity
```

Inheritance

[object](#)  ← [OsmEntity](#) ← Relation

Inherited Members

[OsmEntity.Tags](#) , [OsmEntity.Id](#) , [OsmEntity.Visible](#) , [OsmEntity.Version](#) , [OsmEntity.ChangeSet](#) ,
[OsmEntity.Timestamp](#) , [OsmEntity.User](#) , [OsmEntity.AddTag\(string, string\)](#) , [OsmEntity.RemoveTag\(string\)](#) ,
[OsmEntity.HasTagKey\(string\)](#) , [OsmEntity.HasAnyTagKey\(IEnumerable<string>\)](#) ,
[OsmEntity.HasAllTagKeys\(IEnumerable<string>\)](#) , [OsmEntity.HasTagValue\(string\)](#) ,
[OsmEntity.HasAnyTagValue\(IEnumerable<string>\)](#) , [OsmEntity.HasAllTagValues\(IEnumerable<string>\)](#) ,
[OsmEntity.HasTag\(string, string\)](#) , [OsmEntity.HasAnyTag\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAnyTag\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAnyTag\(IEnumerable<KeyValuePair<string, string>>\)](#) ,
[OsmEntity.HasAllTags\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAllTags\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAllTags\(IEnumerable<KeyValuePair<string, string>>\)](#) , [object.Equals\(object\)](#)  ,
[object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Relation(long, bool, int, long, DateTime, User, IList<Member>)

Initializes a new [Relation](#) object without specified tags.

```
public Relation(long id, bool visible, int version, long changeSet, DateTime timestamp, User  
user, IList<Member> members)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if **members** is **null**.

Relation(long, bool, int, long, DateTime, User, IList<Member>, Dictionary<string, string>?)

Initializes a new [Relation](#) object with specified tags.

```
public Relation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, IList<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**..

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if `members` is `null`.

Relation(long, IList<Member>)

Initializes a new default [Relation](#) object without specified tags.

```
public Relation(long id, IList<Member> members)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`members` [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `members` contain less than one element.

[ArgumentNullException](#)

Throws if `members` is `null`.

Relation(long, IList<Member>, Dictionary<string, string>?)

Initializes a new default [Relation](#) object with specified tags.

```
public Relation(long id, IList<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if **members** is **null**.

Properties

Members

List of the [Relation](#) instance's members.

```
[JsonPropertyName("members")]
[JsonPropertyOrder(6)]
public IList<Member> Members { get; }
```

Property Value

[IList](#) <[Member](#)>

Class ServiceCollectionExtensions

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

```
public static class ServiceCollectionExtensions
```

Inheritance

[object](#)  ← ServiceCollectionExtensions

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Methods

AddOsmToolkit(IServiceCollection)

```
public static IServiceCollection AddOsmToolkit(this IServiceCollection services)
```

Parameters

services [IServiceCollection](#) 

Returns

[IServiceCollection](#) 

Class User

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Contains user data about OSM-creators.

```
public class User
```

Inheritance

[object](#) ← User

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

User(long, string)

Initializes a new [User](#) object.

```
public User(long id, string name)
```

Parameters

id [long](#)

Unique identifier for [User](#).

name [string](#)

Name of the [User](#).

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0.

[ArgumentException](#)

Throws if `name` is `null` or an empty string.

Properties

Id

Unique identifier for [User](#).

```
[JsonPropertyName("id")]  
[JsonPropertyOrder(0)]  
public long Id { get; }
```

Property Value

[long](#)

Name

Name of the [User](#).

```
[JsonPropertyName("name")]  
[JsonPropertyOrder(1)]  
public string Name { get; set; }
```

Property Value

[string](#)

Class Way

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Connection between two or more nodes.

```
public class Way : OsmEntity
```

Inheritance

[object](#)  ← [OsmEntity](#) ← Way

Inherited Members

[OsmEntity.Tags](#) , [OsmEntity.Id](#) , [OsmEntity.Visible](#) , [OsmEntity.Version](#) , [OsmEntity.ChangeSet](#) ,
[OsmEntity.Timestamp](#) , [OsmEntity.User](#) , [OsmEntity.AddTag\(string, string\)](#) , [OsmEntity.RemoveTag\(string\)](#) ,
[OsmEntity.HasTagKey\(string\)](#) , [OsmEntity.HasAnyTagKey\(IEnumerable<string>\)](#) ,
[OsmEntity.HasAllTagKeys\(IEnumerable<string>\)](#) , [OsmEntity.HasTagValue\(string\)](#) ,
[OsmEntity.HasAnyTagValue\(IEnumerable<string>\)](#) , [OsmEntity.HasAllTagValues\(IEnumerable<string>\)](#) ,
[OsmEntity.HasTag\(string, string\)](#) , [OsmEntity.HasAnyTag\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAnyTag\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAnyTag\(IEnumerable<KeyValuePair<string, string>>\)](#) ,
[OsmEntity.HasAllTags\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAllTags\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAllTags\(IEnumerable<KeyValuePair<string, string>>\)](#) , [object.Equals\(object\)](#)  ,
[object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Way(long, bool, int, long, DateTime, User, List<long>)

Initializes a new [Way](#) object without specified tags.

```
public Way(long id, bool visible, int version, long changeSet, DateTime timestamp, User  
user, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if **nodeReferenceIds** is **null**.

Way(long, bool, int, long, DateTime, User, List<long>, Dictionary<string, string>?)

Initializes a new [Way](#) object with specified tags.

```
public Way(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if `nodeReferenceIds` is `null`.

Way(long, List<long>)

Initializes a new default [Way](#) object without specified tags.

```
public Way(long id, List<long> nodeReferenceIds)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`nodeReferenceIds` [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `nodeReferenceIds` contains less than two elements.

[ArgumentNullException](#)

Throws if `nodeReferenceIds` is `null`.

Way(long, List<long>, Dictionary<string, string>?)

Initializes a new default [Way](#) object with specified tags.

```
public Way(long id, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if **nodeReferenceIds** is **null**.

Properties

NodeReferenceIds

A read-only list of reference ids containing the [Way](#)'s nodes.

```
[JsonPropertyName("nodeRefs")]
[JsonPropertyOrder(6)]
public IReadOnlyList<long> NodeReferenceIds { get; }
```

Property Value

[IReadOnlyList](#) <[long](#)>

Methods

IsOneWay()

Checks if the [Way](#) is a oneway or not.

```
public bool IsOneWay()
```

Returns

[bool](#)

`true`, if it is a oneway. Otherwise; `false`.

Namespace OsmToolkit.Factories

Classes

[OsmEntityFactory](#)

Factory class responsible for creating instances of [OsmEntity](#), such as [Node](#), [Way](#) and [Relation](#).

Interfaces

[IOsmEntityFactory](#)

Defines factory methods for creating instances of [OsmEntity](#), such as [Node](#), [Way](#) and [Relation](#).

Interface IOsmEntityFactory

Namespace: [OsmToolkit.Factories](#)

Assembly: OsmToolkit.dll

Defines factory methods for creating instances of [OsmEntity](#), such as [Node](#), [Way](#) and [Relation](#).

```
public interface IOsmEntityFactory
```

Methods

CreateNode(long, bool, int, long, DateTime, User, double, double)

Initializes a new [Node](#) object without specified tags.

```
Node CreateNode(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, double latitude, double longitude)
```

Parameters

id [long](#) 

Unique identifier for [OsmEntity](#).

visible [bool](#) 

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#) 

Current version number of [OsmEntity](#).

changeSet [long](#) 

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#) 

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`..

latitude [double](#)[↗]

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)[↗]

Distance east or west of prime meridian, must be within the range of -180 to 180.

Returns

[Node](#)

A new instance of [Node](#) without specified tags.

CreateNode(long, bool, int, long, DateTime, User, double, double, Dictionary<string, string>?)

Initializes a new [Node](#) object with specified tags.

```
Node CreateNode(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, double latitude, double longitude, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)[↗]

Unique identifier for [OsmEntity](#).

visible [bool](#)[↗]

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

version [int](#)[↗]

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`..

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Node](#)

A new instance of [Node](#) with specified tags.

CreateNode(long, double, double)

Initializes a new base [Node](#) without specified tags.

```
Node CreateNode(long id, double latitude, double longitude)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Returns

[Node](#)

A new default instance of [Node](#) without specified tags.

CreateNode(long, double, double, Dictionary<string, string>?)

Initializes a new base [Node](#) with specified tags.

```
Node CreateNode(long id, double latitude, double longitude, Dictionary<string, string>? tags  
= null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Node](#)

A new default instance of [Node](#) with specified tags.

CreateRelation(long, bool, int, long, DateTime, User, List<Member>)

Initializes a new [Relation](#) object without specified tags.

```
Relation CreateRelation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<Member> members)
```

Parameters

id [long](#) 

Unique identifier for [OsmEntity](#).

visible [bool](#) 

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#) 

Current version number of [OsmEntity](#).

changeSet [long](#) 

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#) 

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

members [List](#)  [Member](#)

Collection of the [Relation](#)'s members.

Returns

[Relation](#)

A new instance of [Relation](#) without specified tags.

CreateRelation(long, bool, int, long, DateTime, User, List<Member>, Dictionary<string, string>?)

Initializes a new [Relation](#) object with specified tags.

```
Relation CreateRelation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`.

members [List](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Relation](#)

A new instance of [Relation](#) with specified tags.

CreateRelation(long, IList<Member>)

Initializes a new default [Relation](#) object without specified tags.

```
Relation CreateRelation(long id, IList<Member> members)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Returns

[Relation](#)

A new default instance of [Relation](#) without specified tags.

CreateRelation(long, IList<Member>, Dictionary<string, string>?)

Initializes a new default [Relation](#) object with specified tags.

```
Relation CreateRelation(long id, IList<Member> members, Dictionary<string, string>? tags)
```

= null)

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Returns

[Relation](#)

A new default instance of [Relation](#) with specified tags.

CreateWay(long, bool, int, long, DateTime, User, List<long>)

Initializes a new [Way](#) object without specified tags.

```
Way CreateWay(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Returns

[Way](#).

A new instance of [Way](#) without specified tags.

CreateWay(long, bool, int, long, DateTime, User, List<long>, Dictionary<string, string>?)

Initializes a new [Way](#) object with specified tags.

```
Way CreateWay(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Way](#)

A new instance of [Way](#) with specified tags.

CreateWay(long, List<long>)

Initializes a new default [Way](#) object without specified tags.

```
Way CreateWay(long id, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Returns

[Way](#).

A new default instance of [Way](#) without specified tags.

CreateWay(long, List<long>, Dictionary<string, string>?)

Initializes a new default [Way](#) object with specified tags.

```
Way CreateWay(long id, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Returns

[Way](#).

A new default instance of [Way](#) with specified tags.

Class OsmEntityFactory

Namespace: [OsmToolkit.Factories](#)

Assembly: OsmToolkit.dll

Factory class responsible for creating instances of [OsmEntity](#), such as [Node](#), [Way](#) and [Relation](#).

```
public class OsmEntityFactory : IOsmEntityFactory
```

Inheritance

[object](#)  ← OsmEntityFactory

Implements

[IOsmEntityFactory](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

OsmEntityFactory()

Initializes a new instance of the [OsmEntityFactory](#) class.

```
public OsmEntityFactory()
```

Methods

CreateNode(long, bool, int, long, DateTime, User, double, double)

Initializes a new [Node](#) object without specified tags.

```
public Node CreateNode(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, double latitude, double longitude)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Returns

[Node](#)

A new instance of [Node](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `latitude` is out of the range -90 to 90 or `longitude` is out of the range -180 to 180.

CreateNode(long, bool, int, long, DateTime, User, double, double, Dictionary<string, string>?)

Initializes a new [Node](#) object with specified tags.

```
public Node CreateNode(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, double latitude, double longitude, Dictionary<string, string>? tags = null)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`visible` [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

`version` [int](#)

Current version number of [OsmEntity](#).

`changeSet` [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

`timestamp` [DateTime](#)

Last time [OsmEntity](#) was changed.

`user` [User](#)

Creator of [OsmEntity](#), can be `null`.

`latitude` [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Node](#)

A new instance of [Node](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **latitude** is out of the range -90 to 90 or **longitude** is out of the range -180 to 180.

CreateNode(long, double, double)

Initializes a new base [Node](#) without specified tags.

```
public Node CreateNode(long id, double latitude, double longitude)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Returns

[Node](#)

A new default instance of [Node](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **latitude** is out of the range -90 to 90 or **longitude** is out of the range -180 to 180.

CreateNode(long, double, double, Dictionary<string, string>?)

Initializes a new base [Node](#) with specified tags.

```
public Node CreateNode(long id, double latitude, double longitude, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Returns

[Node](#)

A new default instance of [Node](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `latitude` is out of the range -90 to 90 or `longitude` is out of the range -180 to 180.

CreateRelation(long, bool, int, long, DateTime, User, List<Member>)

Initializes a new [Relation](#) object without specified tags.

```
public Relation CreateRelation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<Member> members)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`visible` [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

`version` [int](#)

Current version number of [OsmEntity](#).

`changeSet` [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

`timestamp` [DateTime](#)

Last time [OsmEntity](#) was changed.

`user` [User](#)

Creator of [OsmEntity](#), can be `null`.

members [List](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Returns

[Relation](#)

A new instance of [Relation](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if **members** is **null**.

CreateRelation(long, bool, int, long, DateTime, User, List<Member>, Dictionary<string, string>?)

Initializes a new [Relation](#) object with specified tags.

```
public Relation CreateRelation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`.

members [List](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Relation](#)

A new instance of [Relation](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `members` contain less than one element.

[ArgumentNullException](#)

Throws if `members` is `null`.

CreateRelation(long, IList<Member>)

Initializes a new default [Relation](#) object without specified tags.

```
public Relation CreateRelation(long id, IList<Member> members)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Returns

[Relation](#)

A new default instance of [Relation](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if **members** is **null**.

CreateRelation(long, IList<Member>, Dictionary<string, string>?)

Initializes a new default [Relation](#) object with specified tags.

```
public Relation CreateRelation(long id, IList<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Returns

[Relation](#)

A new default instance of [Relation](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if **members** is **null**.

CreateWay(long, bool, int, long, DateTime, User, List<long>)

Initializes a new [Way](#) object without specified tags.

```
public Way CreateWay(long id, bool visible, int version, long changeSet, DateTime timestamp,
    User user, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Returns

[Way](#).

A new instance of [Way](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if **nodeReferenceIds** is **null**.

CreateWay(long, bool, int, long, DateTime, User, List<long>, Dictionary<string, string>?)

Initializes a new [Way](#) object with specified tags.

```
public Way CreateWay(long id, bool visible, int version, long changeSet, DateTime timestamp,
    User user, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Returns

[Way](#)

A new instance of [Way](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if `nodeReferenceIds` is `null`.

CreateWay(long, List<long>)

Initializes a new default [Way](#) object without specified tags.

```
public Way CreateWay(long id, List<long> nodeReferenceIds)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`nodeReferenceIds` [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Returns

[Way](#)

A new default instance of [Way](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `nodeReferenceIds` contains less than two elements.

[ArgumentNullException](#)

Throws if `nodeReferenceIds` is `null`.

CreateWay(long, List<long>, Dictionary<string, string>?)

Initializes a new default [Way](#) object with specified tags.

```
public Way CreateWay(long id, List<long> nodeReferenceIds, Dictionary<string, string>? tags  
= null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Way](#)

A new default instance of [Way](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if **nodeReferenceIds** is `null`.

Namespace OsmToolkit.Finders

Classes

[PathOptions](#)

Represents options for path traversal through [Way](#) instances.

Interfaces

[INearestNodesFinder](#)

Defines finder methods for finding the nearest **possible** [Node](#) instances.

[IOsmFinderV2<T>](#)

Defines finder methods for finding specific **T** instances in an [OsmData](#) instance. This interface replaces [IOsmFinder<T>](#) and provides additional features.

[IOsmFinder<T>](#)

Defines finder methods for finding specific **T** instances in an [OsmData](#) instance.

[IOsmValueFinder<T>](#)

Defines finder methods for finding [OsmEntity](#) instances by **tag** data.

[IShortestPathFinder](#)

Defines finder methods for finding the **shortest** possible path.

[IWithinDistanceFinder<T>](#)

Defines finder methods for finding nearby [OsmEntity](#) instances within a specified **radius**.

Enums

[TravelMode](#)

Selects forms of transport or travel to know what type of **highway:*** to filter out using [PathOptions](#).

Follows https://wiki.openstreetmap.org/wiki/OSM_tags_for_routing/Access_restrictions#Norway.[↗]

Interface INearestNodesFinder

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding the nearest **possible** [Node](#) instances.

```
public interface INearestNodesFinder
```

Methods

FindNearbyNodes(OsmData, Node, int)

Finds a limited amount of nearby [Node](#) instances from a specified [Node](#)'s coordinate. Allows nodes from the same [Way](#) and [Relation](#).

```
IReadOnlyList<Node> FindNearbyNodes(OsmData data, Node node, int limit)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

node [Node](#)

Node for [INearestNodesFinder](#) to search from.

limit [int](#) 

Max limit of nearby [Node](#) instances to find.

Returns

[IReadOnlyList](#)  <[Node](#)>

An [IReadOnlyList<T>](#)  instance with a limited amount of [Node](#) instances.

FindNearbyNodes(OsmData, Node, int, Dictionary<string, string>?, bool, bool)

Finds a limited amount of nearby [Node](#) instances from a specified [Node](#)'s coordinate with optional tags and rules for allowing nodes from the [Way](#) and [Relation](#).

```
ICollection<Node> FindNearbyNodes(OsmData data, Node node, int limit, Dictionary<string, string>? tags = null, bool allowSameWay = true, bool allowSameRelation = true)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

node [Node](#)

Node for [INearestNodesFinder](#) to search from.

limit [int](#)

Max limit of nearby [Node](#) instances to find.

tags [Dictionary](#) <[string](#), [string](#)>

Dictionary containing key-value string pairs describing the metadata of the [OsmEntity](#) instances.

allowSameWay [bool](#)

Whether multiple [Node](#) instances within the same [Way](#) can be included in the results.

allowSameRelation [bool](#)

Whether multiple [Node](#) instances within the same [Relation](#) can be included in the results.

Returns

[ICollection](#) <[Node](#)>

An [ICollection<T>](#) instance with a limited amount of [Node](#) instances.

FindNearbyNodes(OsmData, double, double, int)

Finds a limited amount of nearby [Node](#) instances from a specified coordinate. Allows nodes from the same [Way](#) and [Relation](#).

```
ICollection<Node> FindNearbyNodes(OsmData data, double latitude, double longitude,
int limit)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

latitude [double](#)

Latitude for [INearestNodesFinder](#) to search from.

longitude [double](#)

Longitude for [INearestNodesFinder](#) to search from.

limit [int](#)

Max limit of nearby [Node](#) instances to find.

Returns

[ICollection](#) <[Node](#)>

An [ICollection<T>](#) instance with a limited amount of [Node](#) instances.

FindNearbyNodes(OsmData, double, double, int, Dictionary<string, string>?, bool, bool)

Finds a limited amount of nearby [Node](#) instances from a specified coordinate with optional tags and rules for allowing nodes from the [Way](#) and [Relation](#).

```
ICollection<Node> FindNearbyNodes(OsmData data, double latitude, double longitude, int
limit, Dictionary<string, string>? tags = null, bool allowSameWay = true, bool
allowSameRelation = true)
```


Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

latitude [double](#) 

Latitude for [INearestNodesFinder](#) to search from.

longitude [double](#) 

Longitude for [INearestNodesFinder](#) to search from.

limit [int](#) 

Max limit of nearby [Node](#) instances to find.

tags [Dictionary](#)  [string](#) , [string](#)  >

Dictionary containing key-value string pairs describing the metadata of the [OsmEntity](#) instances.

allowSameWay [bool](#) 

Whether multiple [Node](#) instances within the same [Way](#) can be included in the results.

allowSameRelation [bool](#) 

Whether multiple [Node](#) instances within the same [Relation](#) can be included in the results.

Returns

[IReadOnlyList](#)  [Node](#) 

An [IReadOnlyList<T>](#)  instance with a limited amount of [Node](#) instances.

FindNearestNode(OsmData, Node)

Finds the nearest possible [Node](#) instance from a specified [Node](#)'s coordinate.

```
Node? FindNearestNode(OsmData data, Node node)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

node [Node](#)

Node for [INearestNodesFinder](#) to search from.

Returns

[Node](#)

Nearest possible [Node](#) instance, otherwise; `null`.

FindNearestNode(OsmData, Node, Dictionary<string, string>)

Finds the nearest possible [Node](#) instance from a specified [Node](#)'s coordinate with a tags filter.

```
Node? FindNearestNode(OsmData data, Node node, Dictionary<string, string> tags)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

node [Node](#)

Node for [INearestNodesFinder](#) to search from.

tags [Dictionary](#) <[string](#), [string](#)>

Dictionary containing key-value string pairs describing the metadata of the [OsmEntity](#) instances.

Returns

[Node](#)

Nearest possible [Node](#) instance, otherwise; `null`.

FindNearestNode(OsmData, double, double)

Finds the nearest possible [Node](#) instance from a specified coordinate.

```
Node? FindNearestNode(OsmData data, double latitude, double longitude)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

latitude [double](#) 

Latitude for [INearestNodesFinder](#) to search from.

longitude [double](#) 

Longitude for [INearestNodesFinder](#) to search from.

Returns

[Node](#)

Nearest possible [Node](#) instance, otherwise; **null**.

FindNearestNode(OsmData, double, double, Dictionary<string, string>)

Finds the nearest possible [Node](#) instance from a specified coordinate with a tags filter.

```
Node? FindNearestNode(OsmData data, double latitude, double longitude, Dictionary<string, string> tags)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

latitude [double](#) 

Latitude for [INearestNodesFinder](#) to search from.

longitude [double](#)

Longitude for [INearestNodesFinder](#) to search from.

tags [Dictionary](#) <[string](#), [string](#)>

Dictionary containing key-value string pairs describing the metadata of the [OsmEntity](#) instances.

Returns

[Node](#)

Nearest possible [Node](#) instance, otherwise; `null`.

Interface IOsmFinderV2<T>

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding specific **T** instances in an [OsmData](#) instance. This interface replaces [IOsmFinder<T>](#) and provides additional features.

```
public interface IOsmFinderV2<T> : IOsmValueFinder<T>, INearestNodesFinder,
    IWithinDistanceFinder<T>, IShortestPathFinder where T : OsmEntity
```

Type Parameters

T

Type of [OsmEntity](#) instances to find.

Inherited Members

[IOsmValueFinder<T>.FindByOsmId\(OsmData, long\)](#), [IOsmValueFinder<T>.FindByTag\(OsmData, string\)](#),
[IOsmValueFinder<T>.FindByTag\(OsmData, string, string\)](#),
[IOsmValueFinder<T>.FindByTags\(OsmData, Dictionary<string, string>\)](#),
[INearestNodesFinder.FindNearestNode\(OsmData, Node\)](#),
[INearestNodesFinder.FindNearestNode\(OsmData, Node, Dictionary<string, string>\)](#),
[INearestNodesFinder.FindNearestNode\(OsmData, double, double\)](#),
[INearestNodesFinder.FindNearestNode\(OsmData, double, double, Dictionary<string, string>\)](#),
[INearestNodesFinder.FindNearbyNodes\(OsmData, Node, int\)](#),
[INearestNodesFinder.FindNearbyNodes\(OsmData, Node, int, Dictionary<string, string>, bool, bool\)](#),
[INearestNodesFinder.FindNearbyNodes\(OsmData, double, double, int\)](#),
[INearestNodesFinder.FindNearbyNodes\(OsmData, double, double, int, Dictionary<string, string>, bool, bool\)](#),
[IWithinDistanceFinder<T>.FindNearByRadius\(OsmData, Node, double\)](#),
[IWithinDistanceFinder<T>.FindNearByRadius\(OsmData, double, double, double\)](#),
[IWithinDistanceFinder<T>.FindNearByPathDistance\(OsmData, Node, double\)](#),
[IWithinDistanceFinder<T>.FindNearByPathDistance\(OsmData, double, double, double\)](#),
[IShortestPathFinder.FindShortestPath\(OsmData, Node, Node\)](#),
[IShortestPathFinder.FindShortestPath\(OsmData, Node, Node, PathOptions\)](#),
[IShortestPathFinder.FindShortestPath\(OsmData, double, double, double, double\)](#),
[IShortestPathFinder.FindShortestPath\(OsmData, double, double, double, double, PathOptions\)](#)

Interface IOsmFinder<T>

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding specific **T** instances in an [OsmData](#) instance.

```
[Obsolete("Use IOsmFinderV2 instead.", false)]  
public interface IOsmFinder<T> where T : OsmEntity
```

Type Parameters

T

Type of [OsmEntity](#) instances to find.

Methods

FindByTag(OsmData, string)

Finds **T** instances that contain a specified tag by only key.

```
OsmData FindByTag(OsmData data, string key)
```

Parameters

data [OsmData](#)

Data for [IOsmFinder<T>](#) to use and search through.

key [string](#) 

The key value to access the corresponding metadata, cannot be **null** or an empty string.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tag.

FindByTag(OsmData, string, string?)

Finds **T** instances that contain a specified tag by key and value.

```
OsmData FindByTag(OsmData data, string key, string? value = null)
```

Parameters

data [OsmData](#)

Data for [IOsmFinder<T>](#) to use and search through.

key [string](#) 

The key value to access the corresponding metadata, cannot be **null** or an empty string.

value [string](#) 

The value describing the corresponding metadata, can be **null**.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tag and value.

FindNearByPathDistance(OsmData, double, double, double)

Finds **T** instances from a specified coordinate that are within a specified path distance.

```
OsmData FindNearByPathDistance(OsmData data, double lat, double lon, double distanceMeters)
```

Parameters

data [OsmData](#)

Data for [IOsmFinder<T>](#) to use and search through.

lat [double](#) 

Latitude for [IOsmFinder<T>](#) to search from.

lon [double](#)

Longitude for [IOsmFinder<T>](#) to search from.

distanceMeters [double](#)

Distance in meters for [IOsmFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within path distance.

FindNearByRadius(OsmData, double, double, double)

Finds **T** instances from a specified coordinate that are within a specified radius.

```
OsmData FindNearByRadius(OsmData data, double lat, double lon, double radiusMeters)
```

Parameters

data [OsmData](#)

Data for [IOsmFinder<T>](#) to use and search through.

lat [double](#)

Latitude for [IOsmFinder<T>](#) to search from.

lon [double](#)

Longitude for [IOsmFinder<T>](#) to search from.

radiusMeters [double](#)

Radius in meters for [IOsmFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within radius.

Interface IOsmValueFinder<T>

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding [OsmEntity](#) instances by **tag** data.

```
public interface IOsmValueFinder<T> where T : OsmEntity
```

Type Parameters

T

Type of [OsmEntity](#) instances to find.

Methods

FindByOsmId(OsmData, long)

Finds **T** instances that have a specified id.

```
T? FindByOsmId(OsmData data, long id)
```

Parameters

data [OsmData](#)

Data for [IOsmValueFinder<T>](#) to use and search through.

id [long](#) 

Unique identifier for **T**, must be of values above 0.

Returns

T

An [OsmEntity](#) instance that have the specified id if found, otherwise **null**.

FindByTag(OsmData, string)

Finds **T** instances that contain a specified tag by only key.

```
OsmData FindByTag(OsmData data, string key)
```

Parameters

data [OsmData](#)

Data for [IOsmValueFinder<T>](#) to use and search through.

key [string](#) 

The key value to access the corresponding metadata, cannot be **null** or an empty string.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tag.

FindByTag(OsmData, string, string?)

Finds **T** instances that contain a specified tag by key and value.

```
OsmData FindByTag(OsmData data, string key, string? value = null)
```

Parameters

data [OsmData](#)

Data for [IOsmValueFinder<T>](#) to use and search through.

key [string](#) 

The key value to access the corresponding metadata, cannot be **null** or an empty string.

value [string](#) 

The value describing the corresponding metadata, can be **null**.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tag and value.

FindByTags(OsmData, Dictionary<string, string>)

Finds **T** instances that contain the specified tags in a dictionary with key-value string pairs.

```
OsmData FindByTags(OsmData data, Dictionary<string, string> tags)
```

Parameters

data [OsmData](#)

Data for [IOsmValueFinder<T>](#) to use and search through.

tags [Dictionary](#) <[string](#), [string](#)>

Dictionary containing key-value string pairs describing the metadata of the **T** instances.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tags and values.

Interface IShortestPathFinder

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding the **shortest** possible path.

```
public interface IShortestPathFinder
```

Methods

FindShortestPath(OsmData, Node, Node)

Finds the shortest possible path made up of [OsmEntity](#) instances from a specified **start** [Node](#)'s coordinate to a specified **target** [Node](#)'s coordinate with **default** [PathOptions](#).

```
OsmPath FindShortestPath(OsmData data, Node startNode, Node targetNode)
```

Parameters

data [OsmData](#)

Data for [IShortestPathFinder](#) to use and search through.

startNode [Node](#)

Node for [IShortestPathFinder](#) to start searching from.

targetNode [Node](#)

Node for [IShortestPathFinder](#) to try to reach.

Returns

[OsmPath](#)

An [OsmPath](#) instance with an [OsmData](#) instance containing the [OsmEntity](#) instances that make up the shortest valid path, the total distance in meters and **startnode** and **endnode** of the path. If no valid path

was found, an empty [OsmPath](#) instance is returned with a description describing that it could not find a valid path.

FindShortestPath(OsmData, Node, Node, PathOptions)

Finds the shortest possible path made up of [OsmEntity](#) instances from a specified **start** [Node](#)'s coordinate to a specified **target** [Node](#)'s coordinate with **custom** [PathOptions](#).

```
OsmPath FindShortestPath(OsmData data, Node startNode, Node targetNode,  
PathOptions pathOptions)
```

Parameters

data [OsmData](#)

Data for [IShortestPathFinder](#) to use and search through.

startNode [Node](#)

Node for [IShortestPathFinder](#) to start searching from.

targetNode [Node](#)

Node for [IShortestPathFinder](#) to try to reach.

pathOptions [PathOptions](#)

Options defining which [Way](#) instances [IShortestPathFinder](#) can search through.

Returns

[OsmPath](#)

An [OsmPath](#) instance with an [OsmData](#) instance containing the [OsmEntity](#) instances that make up the shortest valid path, the total distance in meters and **startnode** and **endnode** of the path. If no valid path was found, an empty [OsmPath](#) instance is returned with a description describing that it could not find a valid path.

FindShortestPath(OsmData, double, double, double, double)

Finds the shortest possible path made up of [OsmEntity](#) instances from a specified **start** coordinate to a specified **target** coordinate with default [PathOptions](#).

```
OsmPath FindShortestPath(OsmData data, double startLatitude, double startLongitude, double targetLatitude, double targetLongitude)
```

Parameters

data [OsmData](#)

Data for [IShortestPathFinder](#) to use and search through.

startLatitude [double](#)

Latitude for [IShortestPathFinder](#) to start searching from.

startLongitude [double](#)

Longitude for [IShortestPathFinder](#) to start searching from.

targetLatitude [double](#)

Latitude for [IShortestPathFinder](#) to try to reach.

targetLongitude [double](#)

Longitude for [IShortestPathFinder](#) to try to reach.

Returns

[OsmPath](#)

An [OsmPath](#) instance with an [OsmData](#) instance containing the [OsmEntity](#) instances that make up the shortest valid path, the total distance in meters and **startnode** and **endnode** of the path. If no valid path was found, an empty [OsmPath](#) instance is returned with a description describing that it could not find a valid path.

FindShortestPath(OsmData, double, double, double, double, PathOptions)

Finds the shortest possible path made up of [OsmEntity](#) instances from a specified **start** coordinate to a specified **target** coordinate with custom [PathOptions](#).

```
OsmPath FindShortestPath(OsmData data, double startLatitude, double startLongitude, double targetLatitude, double targetLongitude, PathOptions pathOptions)
```

Parameters

data [OsmData](#)

Data for [IShortestPathFinder](#) to use and search through.

startLatitude [double](#)[↗]

Latitude for [IShortestPathFinder](#) to start searching from.

startLongitude [double](#)[↗]

Longitude for [IShortestPathFinder](#) to start searching from.

targetLatitude [double](#)[↗]

Latitude for [IShortestPathFinder](#) to try to reach.

targetLongitude [double](#)[↗]

Longitude for [IShortestPathFinder](#) to try to reach.

pathOptions [PathOptions](#)

Options defining which [Way](#) instances that [OsmData](#) can search through.

Returns

[OsmPath](#)

An [OsmPath](#) instance with an [OsmData](#) instance containing the [OsmEntity](#) instances that make up the shortest valid path, the total distance in meters and **startnode** and **endnode** of the path. If no valid path was found, an empty [OsmPath](#) instance is returned with a description describing that it could not find a valid path.

Interface IWithinDistanceFinder<T>

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding nearby [OsmEntity](#) instances within a specified **radius**.

```
public interface IWithinDistanceFinder<T> where T : OsmEntity
```

Type Parameters

T

Type of [OsmEntity](#) instances to find.

Methods

FindNearByPathDistance(OsmData, Node, double)

Finds **T** instances from a specified [Node](#)'s coordinate that are within a specified path distance.

```
OsmData FindNearByPathDistance(OsmData data, Node node, double distanceMeters)
```

Parameters

data [OsmData](#)

Data for [IWithinDistanceFinder<T>](#) to use and search through.

node [Node](#)

Node for [IWithinDistanceFinder<T>](#) to search from.

distanceMeters [double](#) 

Distance in meters for [IWithinDistanceFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within path distance.

FindNearByPathDistance(OsmData, double, double, double)

Finds **T** instances from a specified coordinate that are within a specified path distance.

```
OsmData FindNearByPathDistance(OsmData data, double lat, double lon, double distanceMeters)
```

Parameters

data [OsmData](#)

Data for [IWithinDistanceFinder<T>](#) to use and search through.

lat [double](#) 

Latitude for [IWithinDistanceFinder<T>](#) to search from.

lon [double](#) 

Longitude for [IWithinDistanceFinder<T>](#) to search from.

distanceMeters [double](#) 

Distance in meters for [IWithinDistanceFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within path distance.

FindNearByRadius(OsmData, Node, double)

Finds **T** instances from a specified [Node](#)'s that are within a specified radius.

```
OsmData FindNearByRadius(OsmData data, Node node, double radiusMeters)
```

Parameters

data [OsmData](#)

Data for [IWithinDistanceFinder<T>](#) to use and search through.

node [Node](#)

Node for [IWithinDistanceFinder<T>](#) to search from.

radiusMeters [double](#)

Radius in meters for [IWithinDistanceFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within radius.

FindNearByRadius(OsmData, double, double, double)

Finds **T** instances from a specified coordinate that are within a specified radius.

```
OsmData FindNearByRadius(OsmData data, double lat, double lon, double radiusMeters)
```

Parameters

data [OsmData](#)

Data for [IWithinDistanceFinder<T>](#) to use and search through.

lat [double](#)

Latitude for [IWithinDistanceFinder<T>](#) to search from.

lon [double](#)

Longitude for [IWithinDistanceFinder<T>](#) to search from.

radiusMeters [double](#)

Radius in meters for [IWithinDistanceFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within radius.

Class PathOptions

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Represents options for path traversal through [Way](#) instances.

```
public class PathOptions
```

Inheritance

[object](#) ← PathOptions

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

PathOptions()

Initializes a new [PathOptions](#) object with **default** options.

```
public PathOptions()
```

PathOptions(TravelMode, bool)

Initializes a new [PathOptions](#) object.

```
public PathOptions(TravelMode mode, bool avoidMotorway)
```

Parameters

mode [TravelMode](#)

Form of transport or travel to determine the paths that can be traversed.

avoidMotorway [bool](#)

Whether to avoid motorways or not.

Properties

AvoidMotorway

Whether to avoid motorways or not, with **false** as default on initialize.

```
public bool AvoidMotorway { get; init; }
```

Property Value

[bool](#)

Mode

[TravelMode](#) to determine which paths can be traversed, with **Any** as default on initialize.

```
public TravelMode Mode { get; init; }
```

Property Value

[TravelMode](#)

Enum TravelMode

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Selects forms of transport or travel to know what type of **highway:*** to filter out using [PathOptions](#).
Follows https://wiki.openstreetmap.org/wiki/OSM_tags_for_routing/Access_restrictions#Norway.[↗]

```
public enum TravelMode
```

Fields

Any = 0

The default travel mode with no specific filters. (does allow pathing in opposite of one-way)

Bicycle = 2

Travel by bicycle adds filters to not path along disallowed paths.

Foot = 1

Travel by foot adds filters to not path along disallowed paths. (does allow pathing in opposite of one-way)

Moped = 3

Travel by moped adds filters to not path along disallowed paths.

MotorCar = 4

Travel by motorcar adds filters to not path along disallowed paths.

Namespace OsmToolkit.Serialization

Interfaces

[IOsmDeserializer](#)

Defines methods for deserializing OSM data into [OsmData](#) objects.

[IOsmSerializer](#)

Defines methods for serializing [OsmData](#) objects into OSM data source.

Interface IOsmDeserializer

Namespace: [OsmToolkit.Serialization](#)

Assembly: OsmToolkit.dll

Defines methods for deserializing OSM data into [OsmData](#) objects.

```
public interface IOsmDeserializer
```

Methods

DeserializeAsync(string, CancellationToken)

Asynchronously deserializes OSM content from a string.

```
Task<OsmData> DeserializeAsync(string input, CancellationToken cancellationToken = default)
```

Parameters

input [string](#) 

A string containing valid OSM data source.

cancellationToken [CancellationToken](#) 

A cancellation token that can be used to cancel the operation.

Returns

[Task](#)  <[OsmData](#)>

An [OsmData](#) object representing the deserialized OSM data.

DeserializeFromFileAsync(string, CancellationToken)

Asynchronously deserializes OSM data from the specified file path.

```
Task<OsmData> DeserializeFromFileAsync(string path, CancellationToken cancellationToken  
= default)
```

Parameters

path [string](#)

The path to the file to deserialize.

cancellationToken [CancellationToken](#)

A cancellation token that can be used to cancel the operation.

Returns

[Task](#) <[OsmData](#)>

An [OsmData](#) object representing the deserialized OSM data.

Exceptions

[ArgumentException](#)

Thrown when the file extension is not 'osm' or 'xml'.

DeserializeFromStreamAsync(Stream, CancellationToken)

Asynchronously deserializes OSM data from a stream.

```
Task<OsmData> DeserializeFromStreamAsync(Stream stream, CancellationToken cancellationToken  
= default)
```

Parameters

stream [Stream](#)

A readable [Stream](#) containing OSM data.

cancellationToken [CancellationToken](#)

A cancellation token that can be used to cancel the operation.

Returns

[Task](#)  [<OsmData>](#)

An [OsmData](#) object representing the deserialized OSM data.

Interface IOsmSerializer

Namespace: [OsmToolkit.Serialization](#)

Assembly: OsmToolkit.dll

Defines methods for serializing [OsmData](#) objects into OSM data source.

```
public interface IOsmSerializer
```

Methods

SerializeAsync(OsmData, CancellationToken)

Asynchronously serializes the given [OsmData](#) object to a string.

```
Task<string> SerializeAsync(OsmData data, CancellationToken cancellationToken = default)
```

Parameters

data [OsmData](#)

The OSM data to serialize.

cancellationToken [CancellationToken](#)

A cancellation token that can be used to cancel the operation.

Returns

[Task](#) <[string](#)>

A string containing the serialized OSM data source.

SerializeToFileAsync(OsmData, string, CancellationToken)

Asynchronously serializes the given [OsmData](#) object and writes the data source to a file.

```
Task SerializeToFileAsync(OsmData data, string path, CancellationToken cancellationToken  
= default)
```

Parameters

data [OsmData](#)

The OSM data to serialize.

path [string](#) 

The path to the file to write to.

cancellationToken [CancellationToken](#) 

A cancellation token that can be used to cancel the operation.

Returns

[Task](#) 

SerializeToStreamAsync(OsmData, Stream, CancellationToken)

Asynchronously serializes the given [OsmData](#) object and writes the data source to the provided stream.

```
Task SerializeToStreamAsync(OsmData data, Stream stream, CancellationToken cancellationToken  
= default)
```

Parameters

data [OsmData](#)

The OSM data to serialize.

stream [Stream](#) 

The stream to write the data source output to.

cancellationToken [CancellationToken](#) 

A cancellation token that can be used to cancel the operation.

Returns

[Task](#)

Namespace OsmToolkit.Serialization.Json

Interfaces

[IOsmJsonDeserializer](#)

Represents a deserializer that parses OSM Json data into [OsmData](#) objects.

[IOsmJsonSerializer](#)

Represents a serializer that parses [OsmData](#) objects into OSM Json.

Interface IOsmJsonDeserializer

Namespace: [OsmToolkit.Serialization.Json](#)

Assembly: OsmToolkit.dll

Represents a deserializer that parses OSM Json data into [OsmData](#) objects.

```
public interface IOsmJsonDeserializer : IOsmDeserializer
```

Inherited Members

[IOsmDeserializer.DeserializeAsync\(string, CancellationToken\)](#) ,
[IOsmDeserializer.DeserializeFromStreamAsync\(Stream, CancellationToken\)](#) ,
[IOsmDeserializer.DeserializeFromFileAsync\(string, CancellationToken\)](#)

Interface IOsmJsonSerializer

Namespace: [OsmToolkit.Serialization.Json](#)

Assembly: OsmToolkit.dll

Represents a serializer that parses [OsmData](#) objects into OSM Json.

```
public interface IOsmJsonSerializer : IOsmSerializer
```

Inherited Members

[IOsmSerializer.SerializeAsync\(OsmData, CancellationToken\)](#) ,

[IOsmSerializer.SerializeToFileAsync\(OsmData, string, CancellationToken\)](#) ,

[IOsmSerializer.SerializeToStreamAsync\(OsmData, Stream, CancellationToken\)](#)

Namespace OsmToolkit.Serialization.Xml

Interfaces

[IOsmXmlDeserializer](#)

Represents a deserializer that parses OSM XML data into [OsmData](#) objects.

[IOsmXmlSerializer](#)

Represents a serializer that parses [OsmData](#) objects into OSM XML.

Interface IOsmXmlDeserializer

Namespace: [OsmToolkit.Serialization.Xml](#)

Assembly: OsmToolkit.dll

Represents a deserializer that parses OSM XML data into [OsmData](#) objects.

```
public interface IOsmXmlDeserializer : IOsmDeserializer
```

Inherited Members

[IOsmDeserializer.DeserializeAsync\(string, CancellationToken\)](#) ,
[IOsmDeserializer.DeserializeFromStreamAsync\(Stream, CancellationToken\)](#) ,
[IOsmDeserializer.DeserializeFromFileAsync\(string, CancellationToken\)](#)

Interface IOsmXmlSerializer

Namespace: [OsmToolkit.Serialization.Xml](#)

Assembly: OsmToolkit.dll

Represents a serializer that parses [OsmData](#) objects into OSM XML.

```
public interface IOsmXmlSerializer : IOsmSerializer
```

Inherited Members

[IOsmSerializer.SerializeAsync\(OsmData, CancellationToken\)](#) ,

[IOsmSerializer.SerializeToFileAsync\(OsmData, string, CancellationToken\)](#) ,

[IOsmSerializer.SerializeToStreamAsync\(OsmData, Stream, CancellationToken\)](#)